Analysis of TP53

Introduction:

TP53 is an essential protein that plays a key role in regulating the cell cycle, DNA repair, and apoptosis. Due to its critical function, it is highly conserved across different species. Studying TP53 sequences from various organisms helps in understanding its evolutionary relationships and identifying conserved functional regions.

Methods:

The TP53 protein sequence was analyzed using BLAST to identify homologous sequences across different species. Multiple sequence alignment was performed using Clustal Omega to detect conserved regions. InterPro was used to identify functional domains within the protein and possible mutations in the protein. MEGA software was used to construct a phylogenetic tree to study evolutionary relationships.

Results:

BLAST analysis showed high sequence similarity between human TP53 and other mammals. The highest sequence identity was observed in closely related primates (100.0%), followed by green monkey (94.57%). The E-values (0.0) indicate highly significant similarity.

Multiple sequence alignment was performed using Clustal Omega using the human sequence (NP_000537.3) as a reference. The results showed high sequence similarity among the studied species. The highest similarity was observed with primates, followed by green monkey, while lower similarity was observed with flying lemur, whale, and lesser Egyptian jerboa. Conserved regions were identified across all sequences, indicating evolutionary conservation of the protein (figure 1).

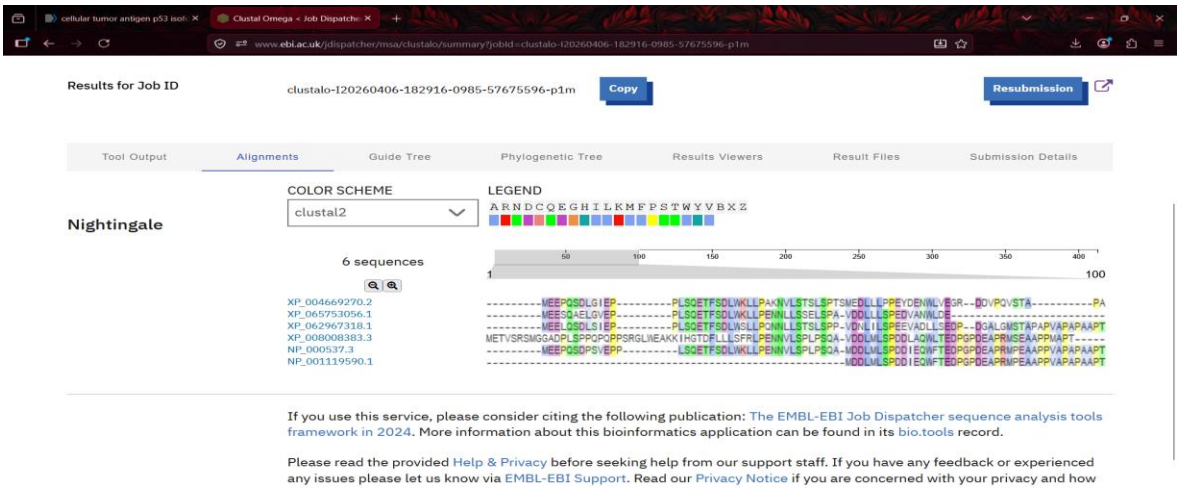

Figure 1: Multiple sequence alignment of the human protein (NP_000537.3) with homologous sequences from different species using Clustal Omega.

InterPro analysis identified several conserved domains, including:

• Transactivation domain

- DNA-binding domain
- Tetramerization domain
- p53-like transcription factor domain

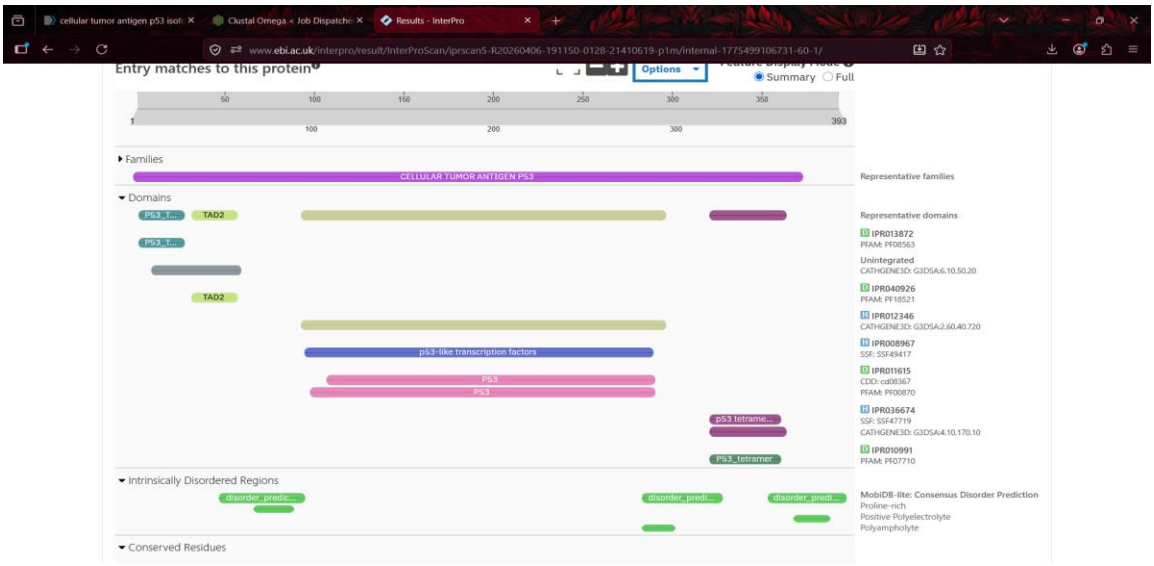

Figure 2: InterPro protein family and domain analysis

A phylogenetic tree was constructed using the Neighbor-Joining method to analyze the evolutionary relationships between TP53 protein sequences across different species.

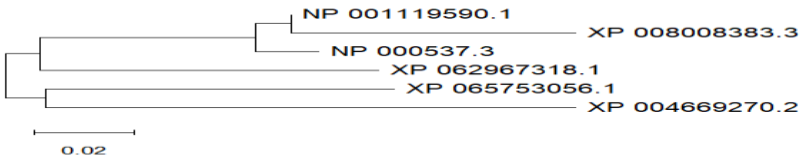

Figure 3: Phylogenetic tree of TP53 protein sequences showing evolutionary divergence. The branching patterns group species based on sequence homology and relative evolutionary distance.

Mutation analysis using MEGA showed some amino acid differences and gaps between the sequences. More variations

were seen in distant species, while conserved regions were present in all sequences, indicating their functional importance.

Figure 4: Mutation analysis showing differences between protein sequences using MEGA software.

Discussion:

The protein sequence was analyzed using BLAST, multiple sequence alignment, InterPro, phylogenetic analysis, and mutation analysis.

The results showed high similarity and conserved regions across species, indicating functional importance. The phylogenetic tree demonstrated a clear evolutionary relationship, with closely related species clustering together. Some variations were observed in distant species, while conserved regions remained unchanged.

Overall, the protein is conserved and functionally important across different species.

Conclusion:

The studied protein is highly conserved across different species, as shown by BLAST, alignment, and phylogenetic analysis. This conservation and evolutionary relationship indicate its important functional role.